CSNS RCS IPM — revised scan matrix (v2), derived from the literature review

Status: complete. 1014 new 100k-particle CSVs in output/v2/ (finished 2026-09-12 05:48 UTC). XMLs from python3 scripts/write_v2_configs.py --write (reuse skipped; extraction ions abort if generation and first-bunch centres differ by more than 1 ns). Re-run with JOBS=4 ./scripts/run_v2.sh (SKIP-if-exists). No v1 CSV was overwritten.

Source. Every change below is traced to a section of LITERATURE_REVIEW.md (the comparison matrix in its Section 5 in particular). The v1 matrices are described in configs/csns_rcs_ipm/emode/README.md and configs/csns_rcs_ipm/imode/README.md.

1. What the review changes

Review finding	Consequence for the matrix	Block
§3.5 Extraction ions are generated 125 ns before the first field-carrying bunch; ext. H_2^+ is a lower bound (+33 % vs ≈ +100 % aligned)	Re-run all extraction ion points with aligned timing (tracking train `LongitudinalOffset` = $-4\sigma_t$ = −80 ns). As-run extraction ion CSVs are kept but superseded.	I1
§3.4 B changes ion widths by < 0.5 % up to 200 G; 0.1 T moves H_2^+ by ≈ 10 %	Drop 200 G from every ion block; keep 0 G and the 0.1 T design field	I1–I3
§3.6 Ion Δx = 5, 10 mm leaves width and centroid unchanged at 100 kW; only the 500 kW aperture pull is new	Drop Δx = +5 mm; keep (10, 0), (0, ±5)	I1
§3.6 Ion size law σ_0^−2 verified on 18 sizes	Size scan reduced to 6 sizes (3, 5, 7, 10, 15, 20 mm)	I1
§3.7 Correction recipe needs h(σ_0, N, V, species) — v1 has only 5 powers ≥ 100 kW	Look-up block: 10 powers from 20 to 500 kW × 5 sizes × 3 species, injection and aligned extraction, 0 G	I2
§3.1 CSNS ions are in the mixed regime (bunch ≈ τ_0); heavy ions at extraction see two bunches	Single-bunch vs 3-bunch train at 100 kW to isolate the multi-bunch term (Shiltsev 1 + 0.8 t_b/τ_0)	I3
§2.3 Δ(B) zeros are spaced by ΔB = π m_e/(e·ToF) ∝ √V to ≤ 5 %	Replace uniform 5 G grids by phase-aware grids: 12 values at k·ΔB (zeros) and (k+½)·ΔB (extrema), k = 1…6, plus 0 and 1000 G — 14 points instead of 61	E3
§2.1 / §2.6 No published low-β benchmark; the N/β column recovers only part of the injection gap	Ramp / β scan: 80, 200, 400, 800, 1200, 1600 MeV at fixed σ = 10 mm and fixed σ_t ∈ {120, 20 ns}, so β is the only variable	E1
§2.4 σ = 3 mm extraction is +1.5 % at 300 G; PRAB fit says 320 G; 0.1 T already on the diagonal	No extra small-beam B fill-in — v1 Block B already has 3–20 mm at 0/100/200/300/1000 G	—
§2.6 CSNS commissioning runs at ≈ 80 kW (IBIC'24); v1 starts at 100 kW	Low-power points 20, 50, 80 kW for the three reference beams at 0/100/200/300/1000 G	E4
Block C at 5 kV: ΔB = 23 G but the grid was 25 G	5 kV dropped from the voltage axis (unresolvable, and outside the CSNS operating range)	E3
§2.1 The 1 % threshold definition ("last excursion") is stricter than the PRAB τ	Analysis change, no runs: report the envelope of Δ(B) (peak amplitude per lobe) alongside the strict threshold	—
§2.6 / §6 Field non-uniformity, MCP, secondary electrons	Blocked: needs the CST/measured cage field map from the hardware group; not counted below	—

2. Axes: v1 versus v2

Axis	v1	v2	Reason
e-mode B grid	0–300 G / 5 G (61), + 1000 G	14 phase-aware values per voltage (E3); 0/100/200/300/500/1000 G (E1)	§2.3 zeros predicted; envelope decays by 300 G
e-mode voltage	5–30 kV / 5 kV on inj. 10 mm (+ 1 kV fine C)	10, 15, 20, 30 kV on ext. 10 mm and inj. 25 mm (25 kV = Block A)	Block C covered one beam only
e-mode power	100–500 kW / 100 kW	+ 20, 50, 80 kW on reference beams	commissioning powers
e-mode beam energy	80 MeV and 1.6 GeV only	6 energies along the ramp, σ_t fixed	low-β benchmark
e-mode size	3–20 mm / 1 mm at 0–300, 1000 G	none — v1 Block B is complete	0.1 T already brings 3 mm onto the diagonal
e-mode offsets	Δy ±10 mm / 1 mm; Δx 5, 10 mm	none	§2.5: < 0.65 % at 300 G, translation invariant
ion B	0, 200, 1000 G	0, 1000 G	§3.4
ion extraction timing	generation 80 ns / first bunch 204.6 ns	aligned (both 80 ns)	§3.5
ion size	3–20 mm / 1 mm	3, 5, 7, 10, 15, 20 mm (extraction only; injection complete)	σ_0^−2 law verified
ion power	100–500 kW	20–500 kW, 10 values (I2, 0 G)	correction look-up
ion voltage	5–30 kV inj. 10 mm	5–30 kV ext. 10 mm, aligned, 0 G	injection complete; extraction superseded
ion offsets	(5,0), (10,0), (0,±5)	(10,0), (0,±5), extraction aligned only	§3.6
ion bunch train	3-bunch circular	+ single bunch (I3)	mixed-regime term

3. Revised blocks

Common settings unchanged from v1: 100 000 secondaries per run, round beams σ_y = σ_x, Voitkiv H_2 DDCS (electrons) / ZeroMomentum ions of 2, 18, 28 u, Boris tracker, N_b ∝ P (7.8 × 10^{12} at 100 kW), uniform E_y = V/231 mm, RNG 1234. SC-off runs do not depend on N and are written once per (beam, σ, V, B); ion SC-off is mass-independent and written for H_2^+ only.

Electron mode

Block	Beam(s)	σ	B [G]	V [kV] / P [kW]	SC-on / SC-off
E1 ramp	80, 200, 400, 800, 1200, 1600 MeV (β = 0.39, 0.57, 0.71, 0.84, 0.90, 0.93); σ_t = 120 ns and 20 ns at every energy	10 mm	0, 100, 200, 300, 500, 1000	25 / 100, 500	144 / 72
E3 phase-aware voltage scan	ext. 1.6 GeV; inj. 80 MeV	10 mm; 25 mm	14 per voltage (table below)	10, 15, 20, 30 / 100, 500	224 / 112
E4 low power	inj. 25 mm; ext. 10 mm; inj. 10 mm	ref.	0, 100, 200, 300, 1000	25 / 20, 50, 80 (SC-off from Block A)	45 / 0

Phase-aware B grids (zeros k·ΔB and extrema (k+½)·ΔB, rounded to 5 G; ΔB = π m_e/(e·ToF), ToF = √(2·115.5 mm·m_e/(e·V/231 mm))):

V	ΔB	Grid [G]
10 kV	32.4 G	0, 15, 30, 50, 65, 80, 95, 115, 130, 145, 160, 180, 195, 1000
15 kV	39.7 G	0, 20, 40, 60, 80, 100, 120, 140, 160, 180, 200, 220, 240, 1000
20 kV	45.9 G	0, 25, 45, 70, 90, 115, 140, 160, 185, 205, 230, 250, 275, 1000
30 kV	56.2 G	0, 30, 55, 85, 110, 140, 170, 195, 225, 255, 280, 310, 335, 1000

Revolution period and bunch spacing for E1 follow from β and the 227.92 m circumference (T_rev = C/βc; spacing = T_rev/2, h = 2). The two σ_t values are deliberately *not* the real ramp values: E1 isolates β at fixed bunch length. A ramp-realistic variant needs the measured bunch-length table (WCM) and is listed under open inputs.

Ion mode

All ion blocks: three species (H_2^+, H_2O^+, N_2^+), 25 kV unless scanned.

Block	Beam(s)	σ	B [G]	Axis scanned	P [kW]	SC-on / SC-off
I1 aligned extraction	ext. 1.6 GeV	3, 5, 7, 10, 15, 20 mm	0, 1000	size	100–500	180 / 12
	ext.	10 mm	0	V = 5, 10, 15, 20, 30 kV	100, 500	30 / 5
	ext.	10 mm	0, 1000	(Δx, Δy) = (10, 0), (0, +5), (0, −5) mm	100, 500	36 / 6
I2 correction look-up	inj. 80 MeV; ext. aligned	5, 10, 15, 20, 25 mm	0	power	20, 50, 80, 100, 150, 200, 250, 300, 400, 500	240 / 6
I3 single bunch	inj. 25; ext. 10; inj. 10 mm	ref.	0	`SingleBunch` tracking beam (`single aligned` at extraction — same $-4\sigma_t$ as generation, not −204.6 ns)	100	9 / 3

I1 and I2 overlap at extraction (sizes 5–20 mm, 100–500 kW, 0 G); overlapping points are counted once, in I1. I3 extraction (10 mm, 100 kW, 0 G) is compared to the I1 aligned 3-bunch point, never to the v1 as-run extraction CSV.

Timing contract (extraction ions)

The v1 artefact is a mismatch of two Virtual-IPM offsets, not a physics effect. Every v2 extraction ion XML must satisfy **generation-window centre = first tracking-bunch centre**. With σ_t = 20 ns that common centre is 80 ns (= $4\sigma_t$).

Beam	Generation (fields off)	Tracking train	First-bunch centre	Status
Injection, v1 and v2	`SingleBunch`, default $-4\sigma_t$ → 480 ns	`CircularBunchTrain`, `LongitudinalOffset` = −spacing/2 = −480 ns	480 ns	already aligned; reuse
Extraction, v1 as-run	`SingleBunch`, default $-4\sigma_t$ → 80 ns	`CircularBunchTrain`, `LongitudinalOffset` = −204.6 ns	204.6 ns	125 ns lag; do not reuse

Extraction, I1 / I2	same 80 ns	`CircularBunchTrain`, `LongitudinalOffset` = −80 ns (= −4σ_t), slug `aligned`	80 ns	the fix
Extraction, I3	same 80 ns	`SingleBunch` with no `LongitudinalOffset` (default −4σ_t). Must not copy `BEAMS["extraction"]["offset_ns"]` = −204.6	80 ns	`train = single aligned`

Acceptance check before any XML is written (generator hook 2): print both centres; abort if they differ by more than 1 ns. The two equivalent fixes of review §3.5 (move the train, or move the generation bunch) are not mixed: v2 always keeps generation at the default −4σ_t and moves the tracking train.

v1 as-run extraction CSVs stay on disk as the mis-timed reference. They are never the baseline for I1–I3, never a reuse hit, and never the look-up column of the §3.7 correction recipe.

4. Run counts (`python3 scripts/scan_matrix_v2.py`)

Block	Description	Runs	Reused	New
E1	ramp / β scan, 6 energies × 2 σ_t, 10 mm, 6 B, 100/500 kW	216	30	186
E3	phase-aware B grid × V (10–30 kV), ext. 10 mm + inj. 25 mm	336	0	336
E4	low power 20/50/80 kW, 3 reference beams, 5 B	45	0	45
I1	aligned extraction: sizes × 5 P × 2 B, V scan, 3 offsets	269	0	269
I2	look-up h(σ_0, N, species): 5 σ × 10 P, inj. + aligned ext.	246	80	166
I3	single bunch vs 3-bunch; extraction is `single aligned`	12	0	12
Total		1124	110	1014

Reused points are v1 runs that coincide exactly (E1: 80 MeV/120 ns and 1.6 GeV/20 ns at Block A fields; I2: injection sizes 5–25 mm at 100–500 kW). For comparison, v1 executed 8127 runs; v2 adds 12 % of that while closing every "partial" and "open" row of the review matrix that simulation can close.

Suggested order. I1 first (it corrects a known bias in published numbers of the report), then I2 (delivers the correction table), E1, E3, E4, I3.

5. Generator hooks required (not implemented)

The v1 generator scripts/generate_csns_configs.py needs five additions before v2 can be written; none of them changes an existing v1 config:

1. **Beam energy and bunch length as parameters** (E1): beam_xml() takes energy, sigma_t_ns, spacing_ns, offset_ns from the BEAMS dict; E1 needs them computed from β (T_rev = 227.92 m / βc) for the six energies and two σ_t values. Suggested family slug ramp_e{energy_mev}mev_s

t{sigma_t_ns}ns_s10x10mm.

2. **Aligned extraction ion timing** (I1, I2, I3): ion_case() today builds the tracking train with circular_train(3, spacing, offset=-spacing/2) (−204.6 ns at extraction). The aligned variant must pass offset = -4·σ_t (−80 ns) so the first bunch centre equals the generation-window centre (80 ns). Abort XML write if the two centres differ by more than 1 ns. New slug infix aligned so as-run and aligned CSVs never collide (csns_extraction_{species}_aligned_s10mm_p100kw_b0G_sc_on.csv). Do not implement the alternative of moving the generation bunch — one convention only.
3. **Single-bunch ion train** (I3): reuse single_bunch_train() (already used for generation, default −4σ_t) for the tracking beam. Do **not** attach BEAMS["extraction"]["offset_ns"] to a SingleBunch — that would recreate the 125 ns lag. Slug infix single (injection) / single_aligned (extraction). Compare I3 extraction to the I1 aligned 3-bunch point, not to the v1 as-run CSV.
4. **New power and field values**: POWERS_KW gains 20, 50, 80 (E4, I2) and 150, 250 (I2); b_tag() already handles arbitrary gauss values, so the phase grids (E3) need only phase_grid_gs(V) from scan_matrix_v2.py.
5. **Evaluators**: evaluate_emode.py / evaluate_imode.py group by family slug; the three new slugs (ramp, aligned, single) need family patterns and a --from-summary replot path like the existing ones. The threshold routine should add the lobe-envelope metric next to the strict "last excursion" threshold.

scan_matrix_v2.py is the single source of the point list; a future --v2 flag of the generator should iterate over output/csns_scan_matrix_v2.csv rather than re-encode the loops.

6. Open inputs (not counted)

Item	Needed from	Unblocks
Cage field map (CST or measured)	CSNS IPM hardware group	e-mode field-non-uniformity block (J-PARC 2× shrink is the reference)
Bunch length $\sigma_t(t)$ along the ramp (WCM)	RCS operations	ramp-realistic E1 variant
Ion initial-momentum model in Virtual-IPM (thermal / dissociation energy)	Virtual-IPM feature check	§6 "ions at rest" assumption; otherwise add ≈ 1 mm in quadrature
MCP window 30 × 80 mm and gain model	IBIC'25 TUPMO41 geometry	detector-level widths at 500 kW / 3 mm (cage-limited today)

7. Execution results

All 1014 new runs finished (4 parallel Virtual-IPM jobs). scripts/evaluate_v2.py pairs each SC-on CSV with its SC-off partner (v2 file, or the v1 file when the off point was reused). 735 SC-on pairs are in output/csns_v2_summary.csv. 78 pairs are unpaired because the matching v1 SC-off particle CSV is not on disk (I2 injection at 20/50/80/150/250 kW; E4 painted injection at 1000 G).

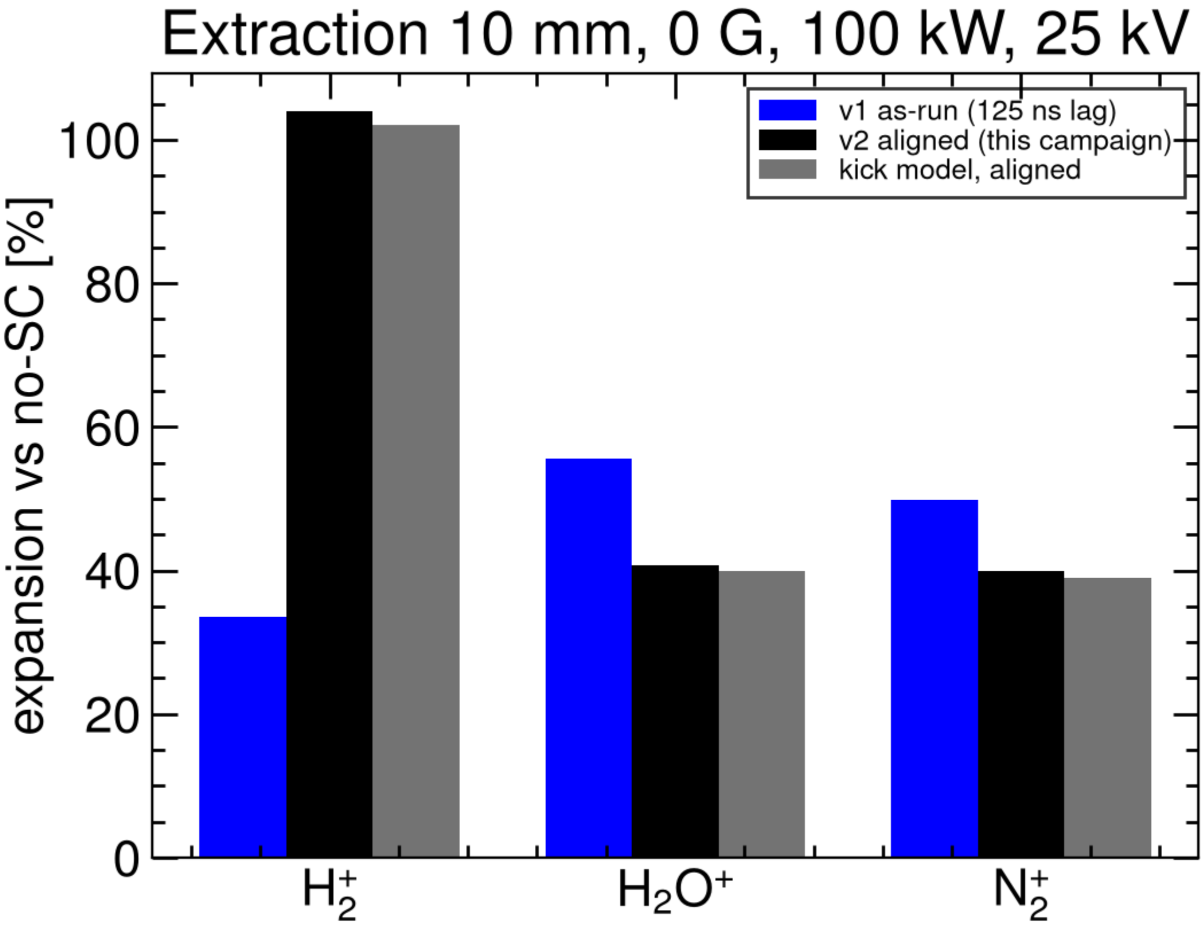

Figure. Extraction 10 mm, 0 G, 100 kW, 25 kV. The v2 aligned Virtual-IPM expansions match the kick-model aligned column of the review (H_2^+ +104 % vs +102 %; H_2O^+ +41 % vs +40 %; N_2^+ +40 % vs +39 %) and replace the v1 as-run lower bound (H_2^+ +33.5 %).

Check	**Result**
I1 timing contract	tracking `LongitudinalOffset` = −80 ns; first bunch at 80 ns
I1 aligned 10 mm / 100 kW / 0 G	H_2^+ +104.1 %, H_2O^+ +40.8 %, N_2^+ +40.0 %
E4 80 kW at 300 G	inj. 25 mm +0.02 %, ext. 10 mm +0.18 %, inj. 10 mm +0.49 %
E1 300 G / 100 kW / σ_t = 120 ns	+0.42 % (200 MeV) → +0.24 % (1.6 GeV); 80 MeV is the reused v1 point

Folded into the combined summary: CSNS_IPM_REPORT.md §4.5 / Figure 14.